

BRAC University (Department of Computer Science and Engineering)
CSE 470 (Software Engineering) | Fall 2023

Quiz 3 (Set A)

Full Marks: 15

Duration: 20 minutes

Name:

Student ID:

Section:

1. (i) Suppose, “ABC Stock Company” wants to let its customers know when the stock prices are going up/ falling down. But it does not want to bother everybody with unwanted notifications. Identify the design pattern needed to implement a notifying system only for customers who are interested in stocks. The notification should look like “Today’s stock price changed to x\$” where x is the current stock price. Complete the given code below. [**10 marks**]

```
public class StockCompany{
    // Add member variables here

    void addCustomer(_____) {
        // Add your code here

    }
    void removeCustomer(_____) {
        // Add your code here

    }
    void notify(){
        // Add your code here

    }
    // Add more functions if needed here

}
```

```
public class Customer{
    // Add your code here

}

public static void main() {
    // Create two customers

    // Create ABC Stock company and make the customers follow it

    // Change the current stock price to 1800$
}
```

(ii) What happens when the current stock price is changed to 1800\$? Explain the codeflow. [**2 marks**]

2. State the differences between Black Box, White Box and Grey Box Testing. [**3 marks**]